In free space, we define the *magnetic flux density* **B** as

$$\mathbf{B} = \mu_0 \mathbf{H} \quad \text{(free space only)}$$

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S} \text{ Wb}$$

webers per square meter (Wb/m^2) or tesla (T).

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

The name given to μ_0 is the *permeability* of free space.

$$\mu = \mu_0 \mu_r$$

$$\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q$$

The magnetic flux lines are closed and do not terminate on a "magnetic charge."

For this reason, Gauss's law for the magnetic field is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

and application of the divergence theorem shows us that

$$\nabla \cdot \mathbf{B} = 0$$

Maxwell's four equations as they apply to static electric fields and steady magnetic fields.

$$\nabla \cdot \mathbf{D} = \rho_v$$
$$\nabla \times \mathbf{E} = 0$$
$$\nabla \times \mathbf{H} = \mathbf{J}$$
$$\nabla \cdot \mathbf{B} = 0$$

Incomplete

Maxwell's four equations as they apply to static electric fields and steady magnetic fields

	Point form	Integral form
1	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dv$
2	$\nabla \times \mathbf{E} = 0$	$\oint \mathbf{E} \cdot d\mathbf{L} = 0$
3	$\nabla \times \mathbf{H} = \mathbf{J}$	$\oint \mathbf{H} \cdot d\mathbf{L} = I = \int_S \mathbf{J} \cdot d\mathbf{S}$
4	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{S} = 0$

Note: For time varying fields, 2 and 3 are different

Scalar and Vector magnetic potentials

- Scalar magnetic potential – $\mathbf{H} = -\nabla V_m$ Similar to $\mathbf{E} = -\nabla V$
- Vector magnetic potential – $\mathbf{B} = \nabla \times \mathbf{A}$
- Reason: Gauss's law for magnetic field is $\nabla \cdot \mathbf{B} = 0$ ---(1)
- WKT vector identity The divergence of the curl of *any* vector field **A** is always zero: --(2)

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

- Comparing (1) and (2) $\mathbf{B} = \nabla \times \mathbf{A}$

Note:

- A problem can be asked – given A find B
- Solution: Find Curl of A to obtain B

Curl Formula in Cylindrical

$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

Curl Formula in Spherical

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

Figure 8.10 shows a boundary between two isotropic homogeneous linear materials with permeabilities μ_1 and μ_2. The boundary condition on the normal components

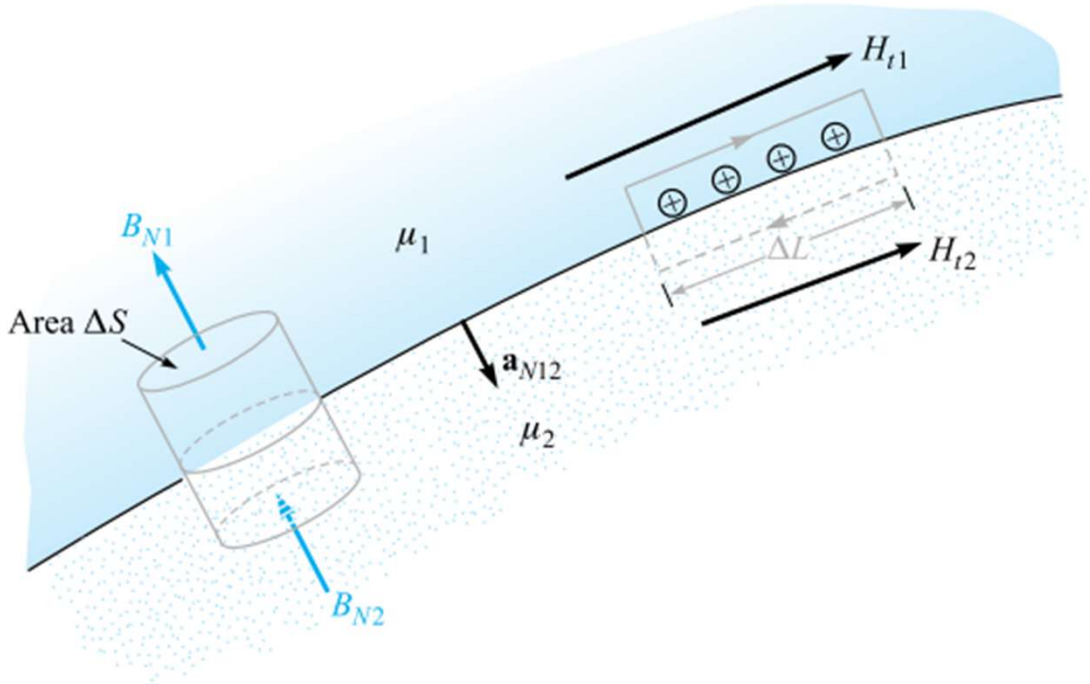
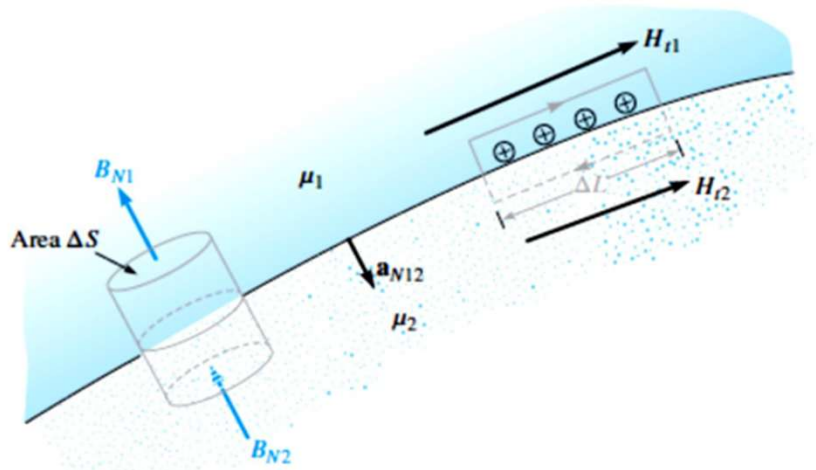


Figure 8.10 A gaussian surface and a closed path are constructed at the boundary between media 1 and 2, having permeabilities of μ_1 and μ_2, respectively. From this we determine the boundary conditions $B_{N1} = B_{N2}$ and $H_{t1} - H_{t2} = K$, the component of the surface current density directed into the page.

MAGNETIC BOUNDARY CONDITIONS



A Gaussian surface and a closed path are constructed at the boundary between media 1 and 2, having permeabilities of μ_1 and μ_2, respectively.

$$\oint_S B \cdot d\mathbf{S} = 0$$

$$B_{N1}\Delta S - B_{N2}\Delta S = 0$$

$$B_{N2} = \mathrm{B}_{N1} \quad \textbf{and} \quad H_{N2} = \frac{\mu_1}{\mu_2} H_{N1}$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

$$H_{t1}\Delta L - H_{t2}\Delta L = K\Delta L$$

$$H_{t1} - H_{t2} = K \quad \textbf{and} \quad \frac{B_{t1}}{\mu_1} - \frac{B_{t2}}{\mu_2} = K$$

$$B_{N2} = \mathrm{B}_{N1}$$

$$(\mathbf{B}_2 - \mathbf{B}_1) \cdot \mathbf{a}_{N12} = 0$$

$$H_{t1} - H_{t2} = K$$

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{N12} = \mathbf{K}$$

Electrostatics

$$E_{\tan 1} = E_{\tan 2}$$

$$D_{N1} - D_{N2} = \rho_S$$

Practice this example from the textbook

EXAMPLE 8.6

To illustrate these relationships with an example, let us assume that $\mu = \mu_1 = 4\ \mu$H/m in region 1 where $z > 0$, whereas $\mu_2 = 7\ \mu$H/m in region 2 wherever $z < 0$. Moreover, let $\mathbf{K} = 80\mathbf{a}_x$ A/m on the surface $z = 0$. We establish a field, $\mathbf{B}_1 = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ mT, in region 1 and seek the value of $\mathbf{B}_2$.

$$\mathbf{B}_{N2} = \mathbf{B}_{N1} = \mathbf{a}_z \text{ mT}$$

$$\mathbf{H}_{t2} = \mathbf{H}_{t1} - \mathbf{a}_{N12} \times \mathbf{K}$$

$$\mathbf{B}_2 = \mathbf{B}_{N2} + \mathbf{B}_{t2} = 3.5\mathbf{a}_x - 4.69\mathbf{a}_y + \mathbf{a}_z \text{mT}$$

Solution. The normal component of $\mathbf{B}_1$ is

$$\mathbf{B}_{N1} = (\mathbf{B}_1 \cdot \mathbf{a}_{N12})\mathbf{a}_{N12} = [(2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z) \cdot (-\mathbf{a}_z)](-\mathbf{a}_z) = \mathbf{a}_z \text{ mT}$$

Thus,

$$\mathbf{B}_{N2} = \mathbf{B}_{N1} = \mathbf{a}_z \text{ mT}$$

We next determine the tangential components:

$$\mathbf{B}_{t1} = \mathbf{B}_1 - \mathbf{B}_{N1} = 2\mathbf{a}_x - 3\mathbf{a}y \text{ mT}$$

and

$$\mathbf{H}_{t1} = \frac{\mathbf{B}_{t1}}{\mu_1} = \frac{(2\mathbf{a}_x - 3\mathbf{a}_y)10^{-3}}{4 \times 10^{-6}} = 500\mathbf{a}_x - 750\mathbf{a}_y \text{ A/m}$$

Thus,

$$\begin{aligned}\mathbf{H}_{t2} &= \mathbf{H}_{t1} - \mathbf{a}_{N12} \times \mathbf{K} = 500\mathbf{a}_x - 750\mathbf{a}_y - (-\mathbf{a}_z) \times 80\mathbf{a}_x \\ &= 500\mathbf{a}_x - 750\mathbf{a}_y + 80\mathbf{a}_y = 500\mathbf{a}_x - 670\mathbf{a}_y \text{ A/m}\end{aligned}$$

and

$$\mathbf{B}_{t2} = \mu_2\mathbf{H}_{t2} = 7 \times 10^{-6}(500\mathbf{a}_x - 670\mathbf{a}_y) = 3.5\mathbf{a}_x - 4.69\mathbf{a}_y \text{ mT}$$

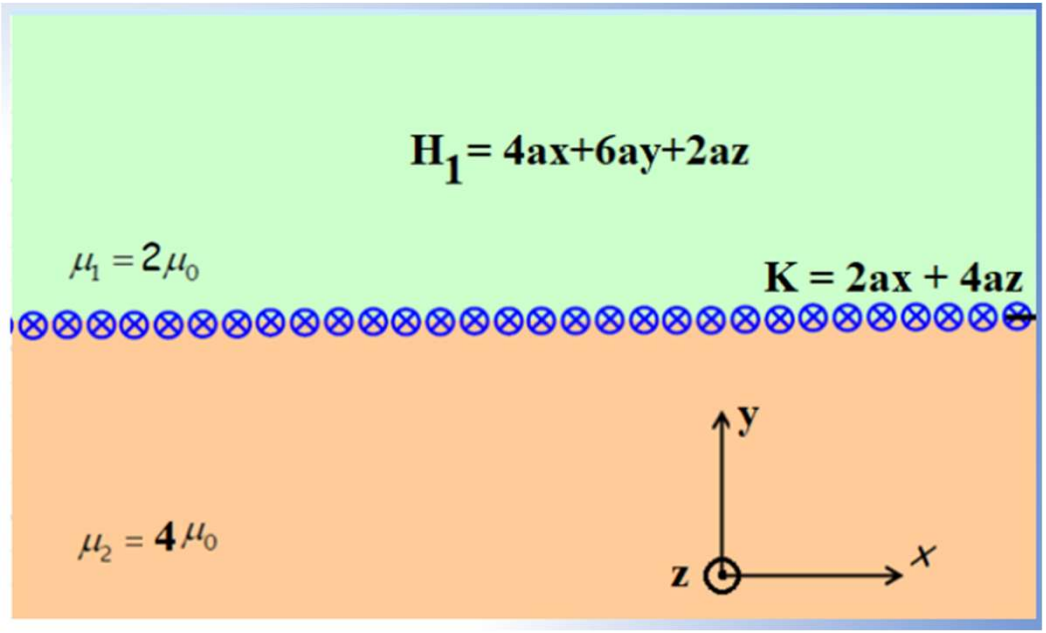
Therefore,

$$\mathbf{B}_2 = \mathbf{B}_{N2} + \mathbf{B}_{t2} = 3.5\mathbf{a}_x - 4.69\mathbf{a}_y + \mathbf{a}_z \text{ mT}$$

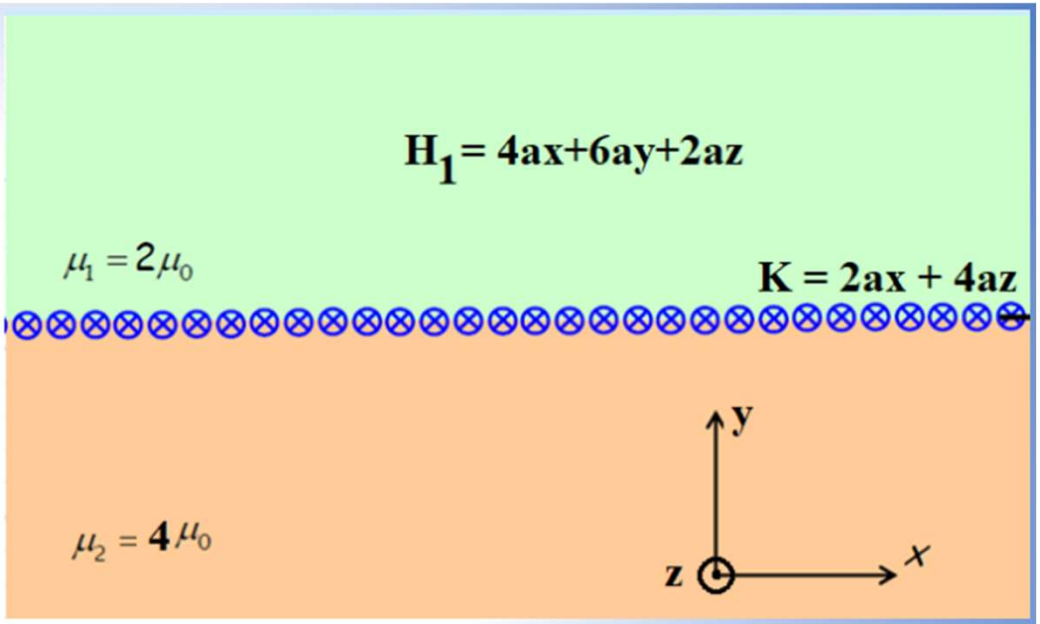
D8.8. Let the permittivity be 5 μH/m in region A where $x < 0$, and 20 μH/m in region B where $x > 0$. If there is a surface current density $\mathbf{K} = 150\mathbf{a}_y - 200\mathbf{a}_z$ A/m at $x = 0$, and if $H_A = 300\mathbf{a}_x - 400\mathbf{a}_y + 500\mathbf{a}_z$ A/m, find: (*a*) $|\mathbf{H}_{tA}|$; (*b*) $|\mathbf{H}_{NA}|$; (*c*) $|\mathbf{H}_{tB}|$; (*d*) $|\mathbf{H}_{NB}|$.

Ans. 640 A/m; 300 A/m; 695 A/m; 75 A/m

Problem on boundary condition:
Find H_2



Problem on boundary condition:
Find H_2



y component is Normal component

x & z components are tengential components

For Normal components, $B_{N2} = B_{N1}$ and $H_{N2} = \frac{\mu_1}{\mu_2} H_{N1}$

For Tengential components, $H_{t1} - H_{t2} = K$

$= \frac{2}{4} \cdot 6\hat{a}_y = 3\hat{a}_y$

$\mathbf{H_1 = 4ax+6ay+2az}$ $\quad K = 2\hat{a}_x + 4\hat{a}_2$

For x component

$H_{t2} = H_{t1} - a_N \times K$ ← ⊥ to H_{t1} with K

$= 4\hat{a}_x - (\hat{a}_y \times 4\hat{a}_2) = 0$

For z component

$H_{t2} = H_{t1} - a_N \times K$ ← ⊥ to H_{t1} with K

$= 2\hat{a}_2 - (a_y \times 2\hat{a}_x)$

$= 4\hat{a}_2$

$H_2 = H_{t2} + H_{N2} = 3\hat{a}_y + 4\hat{a}_2$

Faraday's law

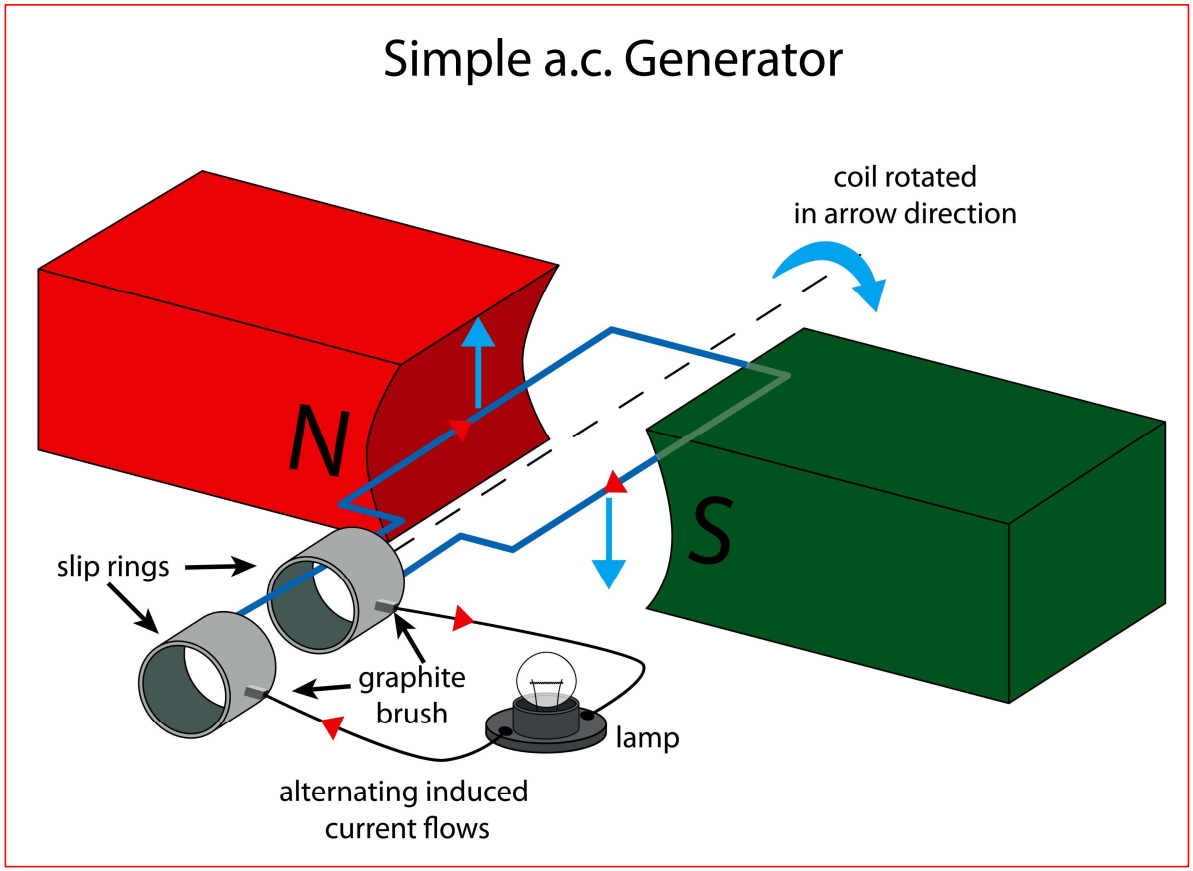
- A time-varying magnetic field produces an *electromotive force* (emf) that may establish a current in a suitable closed circuit.
- An electromotive force is merely a voltage that arises from conductors moving in a magnetic field or from changing magnetic fields
- Faraday's law is customarily stated as

$$\text{emf} = -\frac{d\Phi}{dt}\,\text{V}$$

$$\text{emf} = -N\frac{d\Phi}{dt}$$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L}$$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{S}$$


Simple a.c. Generator
coil rotated
in arrow direction
N
S
slip rings
graphite
brush
lamp
alternating induced
current flows

The key to differentiating the integrals is the **context** (electrostatics vs. electrodynamics) and whether the integral is an **open path** or a **closed loop**.

Integral Expression	Path Type	Physical Meaning	Field Type	Key Principle
$\int_A^B \mathbf{E} \cdot d\mathbf{l}$	**Open Path**	**Potential Difference** ($\Delta V = V_B - V_A$)	Conservative or Non-conservative	Work done moving a charge between two points.
$\oint_C \mathbf{E} \cdot d\mathbf{l}$	**Closed Loop**	**Zero**	**Conservative** ($\mathbf{E}_{\text{conservative}}$)	Kirchhoff's Voltage Law (in static circuits).
$\oint_C \mathbf{E} \cdot d\mathbf{l}$	**Closed Loop**	**Electromotive Force ($\mathcal{E}$)**	**Non-conservative** ($\mathbf{E}_{\text{induced}}$)	**Faraday's Law of Induction.**

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Applying Stokes' theorem

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

This is one of Maxwell's four equations as written in differential, or point, form

Review: Maxwell's four equations as they apply to static electric fields and steady magnetic fields

	Point form	Integral form
1	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dv$
2	$\nabla \times \mathbf{E} = 0$	$\oint \mathbf{E} \cdot d\mathbf{L} = 0$
3	$\nabla \times \mathbf{H} = \mathbf{J}$	$\oint \mathbf{H} \cdot d\mathbf{L} = I = \int_S \mathbf{J} \cdot d\mathbf{S}$
4	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{S} = 0$

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Note: For time varying fields, 2 and 3 are different

DISPLACEMENT CURRENT

- Modifying Ampere's Law for Time-Varying Fields $\nabla \times \mathbf{H} = \mathbf{J}$
- Taking the divergence of each side $\nabla \cdot \nabla \times \mathbf{H} \equiv 0 = \nabla \cdot \mathbf{J}$
- WKT, the equation of current continuity $\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$
- Let $0 = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{G}$ such that $\nabla \cdot \mathbf{G} = \frac{\partial \rho_v}{\partial t}$ Displacement current density G = J_d

Replacing ρ_v with $\nabla \cdot \mathbf{D}$, $\nabla \cdot \mathbf{G} = \frac{\partial}{\partial t}(\nabla \cdot \mathbf{D}) = \nabla \cdot \frac{\partial \mathbf{D}}{\partial t}$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$ Maxwell's equation

Note: $\mathbf{J} = \sigma \mathbf{E}$ $\mathbf{J} = \rho_v \mathbf{v}$ $\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$

Maxwell's equation for time varying fields

Point form		Integral form
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	Surface integral and Stoke's theorem	$\oint \mathbf{E} \cdot d\mathbf{L} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$		$\oint \mathbf{H} \cdot d\mathbf{L} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$
$\nabla \cdot \mathbf{D} = \rho_v$	Volume integral and Divergence theorem	$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \rho_v dv$
$\nabla \cdot \mathbf{B} = 0$		$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$

D9.1. Within a certain region, $\epsilon = 10^{-11}$ F/m and $\mu = 10^{-5}$ H/m. If $B_x = 2 \times 10^{-4} \cos 10^5 t \sin 10^{-3} y$ T: (*a*) use $\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t}$ to find **E**; (*b*) find the total magnetic flux passing through the surface $x = 0, 0 < y < 40$ m, $0 < z < 2$ m, at $t = 1$ μs; (*c*) find the value of the closed line integral of **E** around the perimeter of the given surface.

Ans. (*a*) $-20\,000 \sin 10^5 t \cos 10^{-3} y \mathbf{a}_z$ V/m;

(*b*) 0.318 mWb;

(*c*) −3.19 V

$$\text{emf} = \oint \mathbf{E} \cdot d\mathbf{L} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

A **phasor** is a complex number that represents a sinusoidal function (like a voltage or current that changes with time) in terms of its **magnitude** and **phase angle**.

$$v(t) = V_m \cos(\omega t + \phi)$$

If you have to add or differentiate several of these functions, the trigonometry becomes complex and time-consuming.

The phasor transformation removes the time-dependent part ($\cos(\omega t + \dots)$) and focuses only on the two constants that define the wave:

1. **Magnitude (V_m or V):** How big the wave is.
2. **Phase (ϕ):** Where the wave starts relative to time zero.

The Phasor Transformation

We use Euler's identity to make the transformation: $e^{j\theta} = \cos(\theta) + j\sin(\theta)$.

A time-domain function $v(t)$ is transformed into a frequency-domain phasor $\mathbf{V}$:

$$\begin{aligned} &\text{Time Domain:} \quad v(t) = V_m \cos(\omega t + \phi) \\ &\Downarrow \text{Phasor Transformation} \\ &\text{Frequency Domain:} \quad \mathbf{V} = V_m e^{j\phi} = V_m \angle \phi \end{aligned}$$

Feature	Time Domain ($v(t)$)	Frequency Domain ($\mathbf{V}$, Phasor)
Form	Sinusoidal Function	Complex Number
Variables	Time (t), Frequency (ω)	Magnitude, Phase Angle (ϕ)
Math	Differential equations, Trigonometry	Algebra
Purpose	Shows physical waveform	Simplifies analysis

Maxwell's equations phasor form

$$\mathbf{E} = \mathrm{Re}\left\{\tilde{\mathbf{E}} e^{j\omega t}\right\}$$

$$\mathbf{B} = \mathrm{Re}\left\{\tilde{\mathbf{B}} e^{j\omega t}\right\}$$

$$\nabla \times \left[\mathrm{Re}\left\{\tilde{\mathbf{E}} e^{j\omega t}\right\}\right] = -\frac{\partial}{\partial t}\left[\mathrm{Re}\left\{\tilde{\mathbf{B}} e^{j\omega t}\right\}\right]$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\begin{aligned}\mathrm{Re}\{[\nabla \times \tilde{\mathbf{E}}] e^{j\omega t}\} &= -\mathrm{Re}\left\{\tilde{\mathbf{B}} \frac{\partial}{\partial t} e^{j\omega t}\right\} \\ &= -\mathrm{Re}\left\{\tilde{\mathbf{B}}\, j\omega e^{j\omega t}\right\} \\ &= \mathrm{Re}\left\{[-j\omega \tilde{\mathbf{B}}] e^{j\omega t}\right\}\end{aligned}$$

-
-

$$\nabla \times \tilde{\mathbf{E}} = -j\omega \tilde{\mathbf{B}}$$

Similarly

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \times \tilde{\mathbf{H}} = \tilde{\mathbf{J}} + j\omega \tilde{\mathbf{D}}$$

No change in Gauss's law

Introduction to Uniform Plane Waves

- Let field $\vec{E} = \hat{a}_y E_o \cos(\omega t - \beta z)$ –in free space (No free charges) – y-axis
- β – positive real number
- E is a function of time and space
- We have to find $\vec{B}$, $\vec{D}$ and $\vec{H}$ vectors
- We have to show that $\vec{E}$ and $\vec{H}$ constitutes a wave and propagates in z – direction
- $D = \epsilon_o E \Rightarrow D = \hat{a}_y \epsilon_o E_o \cos(\omega t - \beta z)$
- To find B and H, use Maxwell's equation
- Find curl of E

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

POYNTING'S THEOREM AND WAVE POWER

- Power flow associated with an electromagnetic wave
- Consider Maxwell's curl equation - assume that the medium may be conductive

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

- Take the scalar product of both sides with **E** $\mathbf{E} \cdot \nabla \times \mathbf{H} = \mathbf{E} \cdot \mathbf{J} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}$ **---(1)**

- Introduce the following vector identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mathbf{E} \cdot \nabla \times \mathbf{H} + \mathbf{H} \cdot \nabla \times \mathbf{E}$

- Rewriting eq. (1) → $\mathbf{H} \cdot \nabla \times \mathbf{E} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{J} \cdot \mathbf{E} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}$

- Consider $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

- Therefore $-\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{J} \cdot \mathbf{E} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}$

Rewriting…

$$-\mathbf{H}\cdot\frac{\partial \mathbf{B}}{\partial t}-\nabla\cdot(\mathbf{E}\times\mathbf{H})=\mathbf{J}\cdot\mathbf{E}+\mathbf{E}\cdot\frac{\partial \mathbf{D}}{\partial t}$$

$$-\nabla\cdot(\mathbf{E}\times\mathbf{H})=\mathbf{J}\cdot\mathbf{E}+\epsilon\mathbf{E}\cdot\frac{\partial \mathbf{E}}{\partial t}+\mu\mathbf{H}\cdot\frac{\partial \mathbf{H}}{\partial t}$$

- But

$$\epsilon\mathbf{E}\cdot\frac{\partial \mathbf{E}}{\partial t}=\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\right)$$

(Rearranging the derivatives)

$$\mu\mathbf{H}\cdot\frac{\partial \mathbf{H}}{\partial t}=\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\right)$$

$$-\nabla\cdot(\mathbf{E}\times\mathbf{H})=\mathbf{J}\cdot\mathbf{E}+\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\right)+\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\right)$$

$$-\int_{\text{vol}}\nabla\cdot(\mathbf{E}\times\mathbf{H})\,dv=\int_{\text{vol}}\mathbf{J}\cdot\mathbf{E}\,dv+\int_{\text{vol}}\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\right)dv+\int_{\text{vol}}\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\right)dv$$

$$-\oint_{\text{area}}(\mathbf{E}\times\mathbf{H})\cdot d\mathbf{S}=\int_{\text{vol}}\mathbf{J}\cdot\mathbf{E}\,dv+\frac{d}{dt}\int_{\text{vol}}\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\,dv+\frac{d}{dt}\int_{\text{vol}}\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\,dv$$

Poynting's theorem

$$-\oint_{\text{area}} (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = \int_{\text{vol}} \mathbf{J} \cdot \mathbf{E}\, dv + \frac{d}{dt} \int_{\text{vol}} \frac{1}{2} \mathbf{D} \cdot \mathbf{E}\, dv + \frac{d}{dt} \int_{\text{vol}} \frac{1}{2} \mathbf{B} \cdot \mathbf{H}\, dv$$

Total power flowing *out* of the volume

Ohmic power dissipated within the volume

Total energy stored in the electric field

Total energy stored in the magnetic field

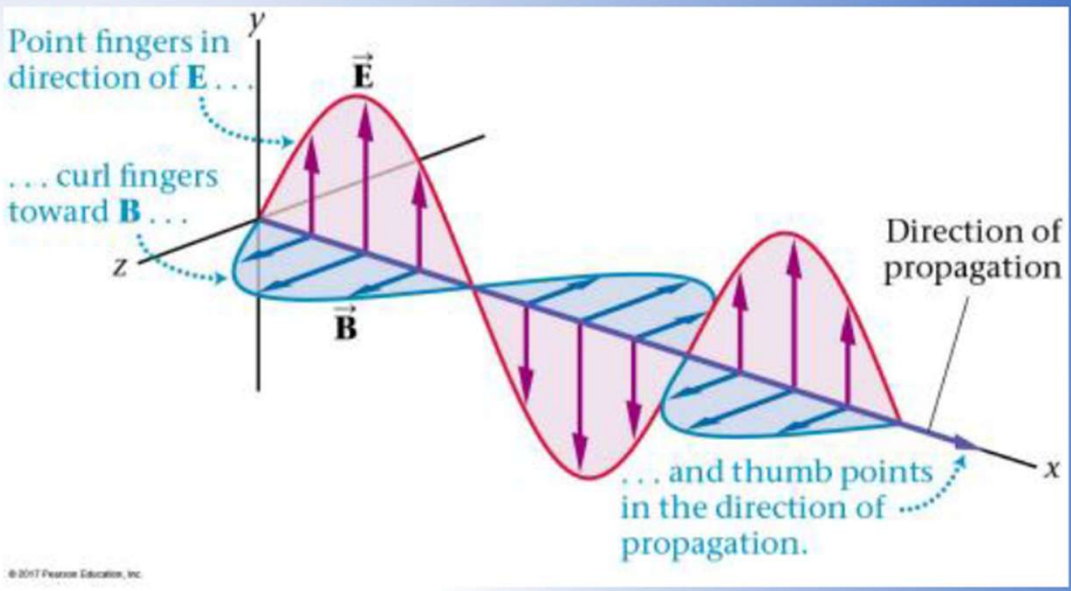
$$\oint_{\text{area}} (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} \quad \text{W}$$

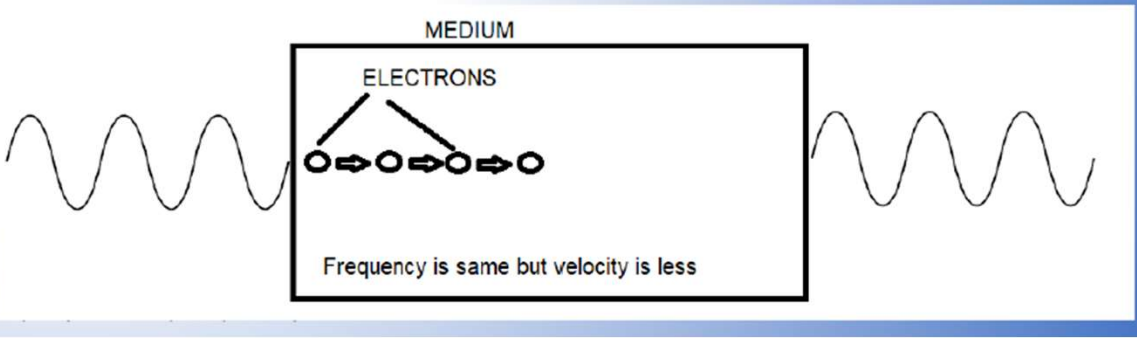
Poynting vector

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad \text{W/m}^2$$

instantaneous power density

- EM waves “propagate” – Source vibrates at a frequency
- This is carried by EM wave – source not required – self propagation
- EM wave don’t require medium for propagation
- If it hits a media – media carries the vibration
- Freq. don’t change, amplitude may change
- Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad \mathrm{W/m^2}$
- Gives instantaneous power density





An **H** field in free space is given as $\mathbf{H}(x,t) = 10\cos(10^8 t - \beta x)\mathbf{a}_y$ A/m.
Find: (a) β; (b) λ; (c) $\mathbf{E}(x,t)$ at $P(0.1, 0.2, 0.3)$ at $t = 1$ ns.

$\beta = \omega/c$, $\beta = 10^8/(3 \times 10^8) = \underline{0.33 \text{ rad/m}}$.

b) λ: We know $\lambda = 2\pi/\beta = \underline{18.9 \text{ m}}$.

$$E(x,t) = -\eta_0 H(x,t) = -3.77 \times 10^3 \cos(10^8 t - \beta x).$$

The vector direction of **E** will be $-\mathbf{a}_z$,

$-a_z \times a_y = a_x$

Direction of propagation is

E X H

At $x = 0.1$, $t = 10^{-9}$ sec,

$$\mathbf{E}(x,t) = -3.77 \times 10^3 \cos[(10^8)(10^{-9}) - (0.33)(0.1)]\mathbf{a}_z$$

$$= -3.77 \times 10^3 \cos(6.7 \times 10^{-2})\mathbf{a}_z$$

$$= \underline{-3.76 \times 10^3 \mathbf{a}_z \text{ V/m}}$$

2. A wave propagating in lossless dielectric has the components E = 500 cos (10^7 t-βz) a_x V/m and H = 1.1 cos (10^7 t-βz) a_y. If the wave is travelling at v = 0.5c, find
a) μ_r b) ε_r c) β d) λ e) η

$$\eta = \sqrt{\frac{\mu}{\varepsilon}}$$

$$V = \frac{1}{\sqrt{\mu\varepsilon}}$$

Given $\omega = 10^7$, $E_0 = 500$ V/m, $H_0 = 1.1$ A/m, $v = 0.5c = 1.5 \times 10^8$ m/s

a) $\eta = \frac{E_0}{H_0} = \sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}} = \frac{500}{1.1} \Rightarrow \frac{\mu_r}{\epsilon_r} = 1.455$

$v = \frac{1}{\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}} = 1.5 \times 10^8 \Rightarrow \mu_r\epsilon_r = 4$

Solving we get
$\mu_r = \sqrt{5.823}$
$\epsilon_r = \sqrt{2.413}$

$$\beta = \omega\sqrt{\mu\,\varepsilon}$$

c) $\beta = \omega\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}$

=0.0645 rad/m

d) $\lambda = \frac{2\pi}{\beta} = 94.2$ m

e) $\eta = \frac{500}{1.1} = 454.545\ \Omega$

A good conductor is planar in form and carries a uniform plane wave that has a wavelength of 0.3 mm and a velocity of 3×10^5 m/s. Assuming the conductor is non-magnetic, determine the frequency and the conductivity:

$$f = \frac{v}{\lambda} = \frac{3 \times 10^5}{3 \times 10^{-4}} = 10^9 \text{ Hz} = \underline{1 \text{ GHz}}$$

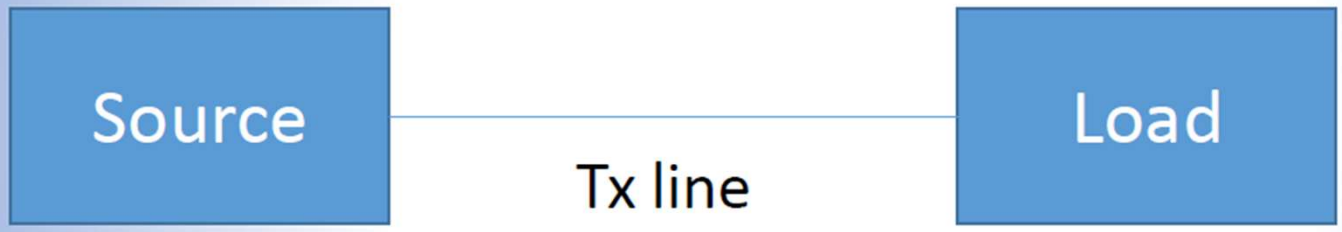
$\delta = \frac{1}{\alpha} = \frac{1}{\beta}$ and $\lambda = \frac{2\pi}{\beta} = 2\pi\delta$

$$\delta = \frac{\lambda}{2\pi} = \frac{1}{\sqrt{\pi f \mu \sigma}} \quad \Rightarrow \quad \sigma = \frac{4\pi}{\lambda^2 f \mu} = \frac{4\pi}{(9 \times 10^{-8})(10^9)(4\pi \times 10^{-7})} = \underline{1.1 \times 10^5 \text{ S/m}}$$

Problem: Write the time domain expression for the electric field of a uniform plane wave in a lossy medium ($\mu_r = 1$) if

$$H = 0.1e^{-200y}\cos(2\pi \times 10^{10}t - 300y)\hat{a}_x\ (A/m)$$

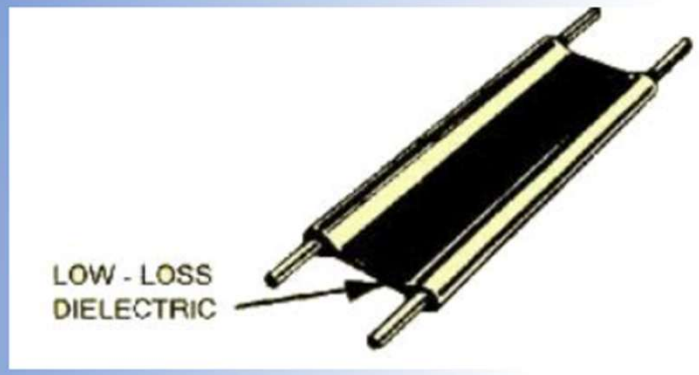
- Transmission lines are used to transmit Electrical Energy or Signals from one point to another
  - Source to destination
  - Generator to load

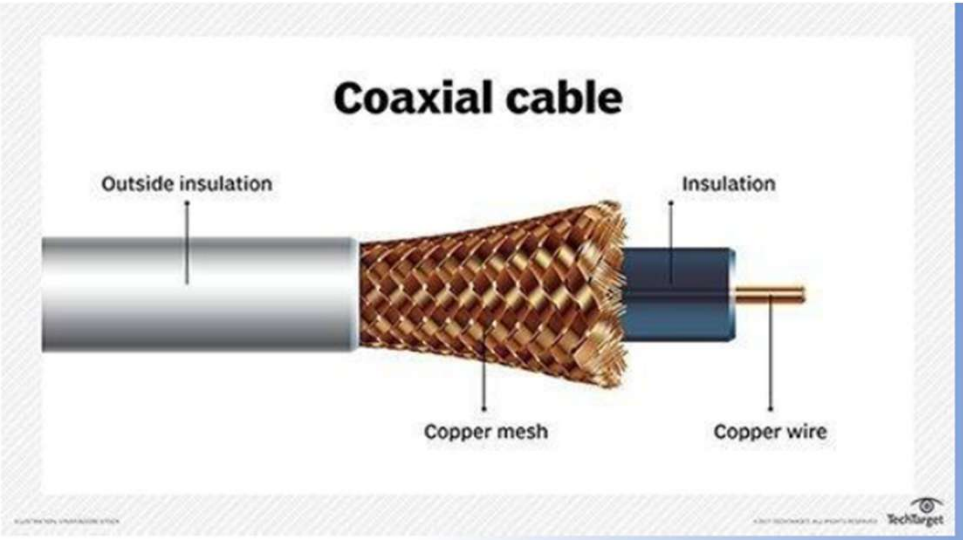


Examples:

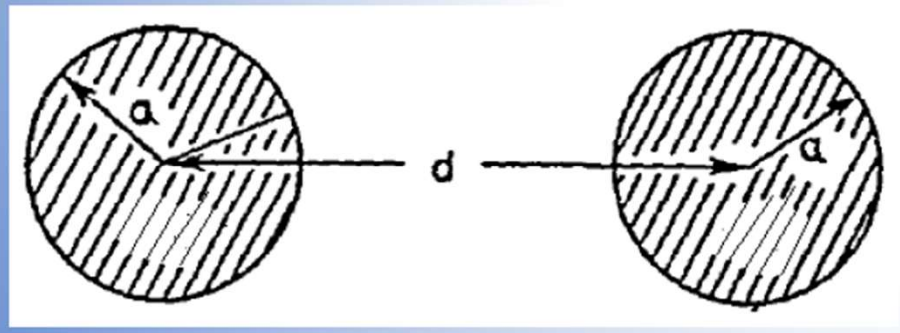
1. Data from one computer to another
2. Signal from Cable TV provider to TV sets
3. Signal from Transmitter to antenna / Antenna to receiver
4. Connections between devices in a RF circuit board

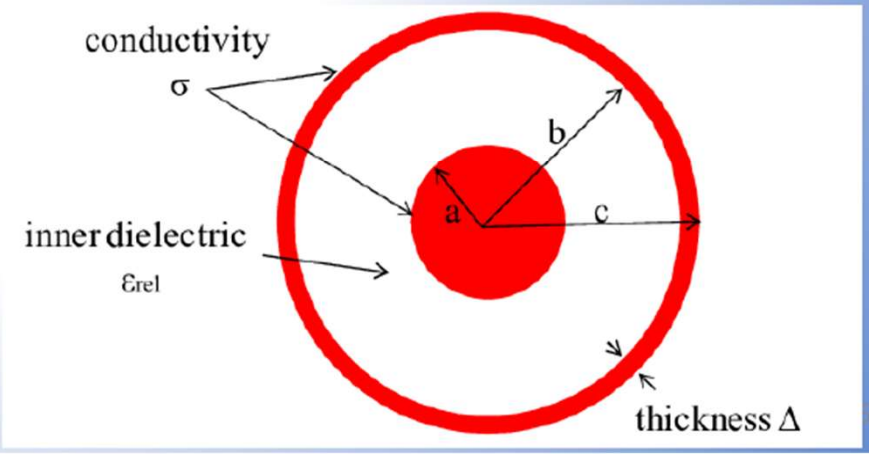
Flat parallel wire





Cross section





Parameters of Transmission Line

- Lumped Parameter
- Distributed Parameter

<u>Lumped Parameter model of TL:</u>

- If the operating wavelength is very large compared to its physical length – lumped circuit element

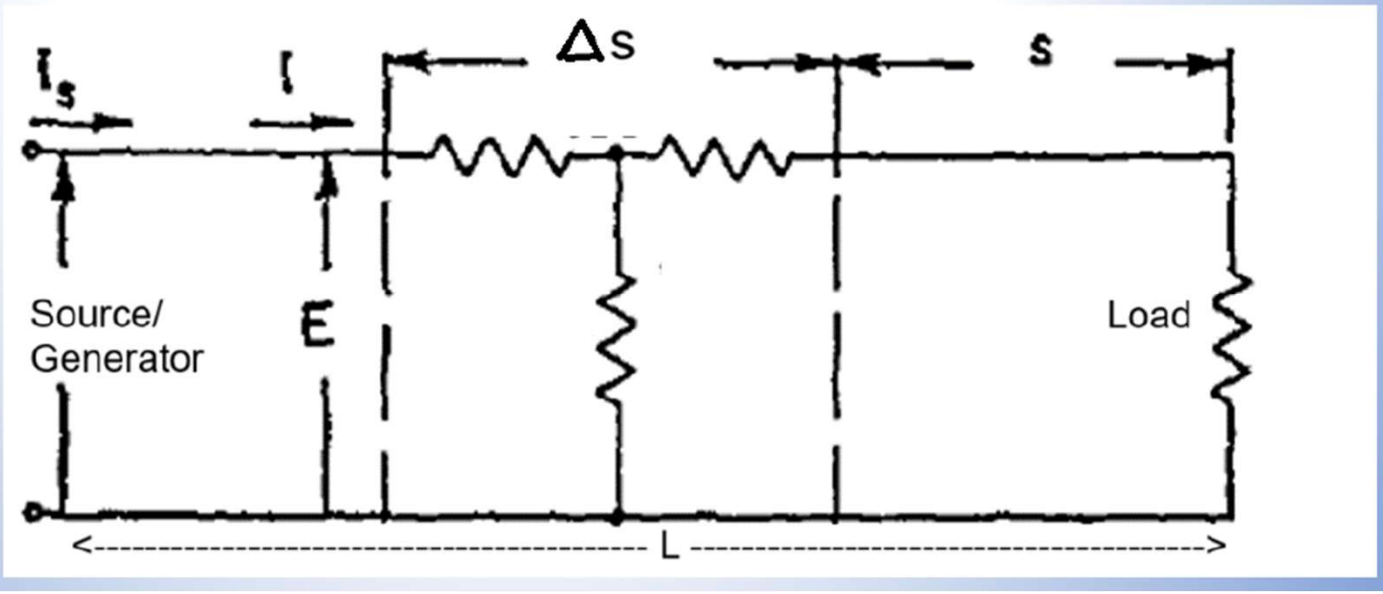
$\lambda >> L$

- For f=300kHz

$\lambda = 1km$

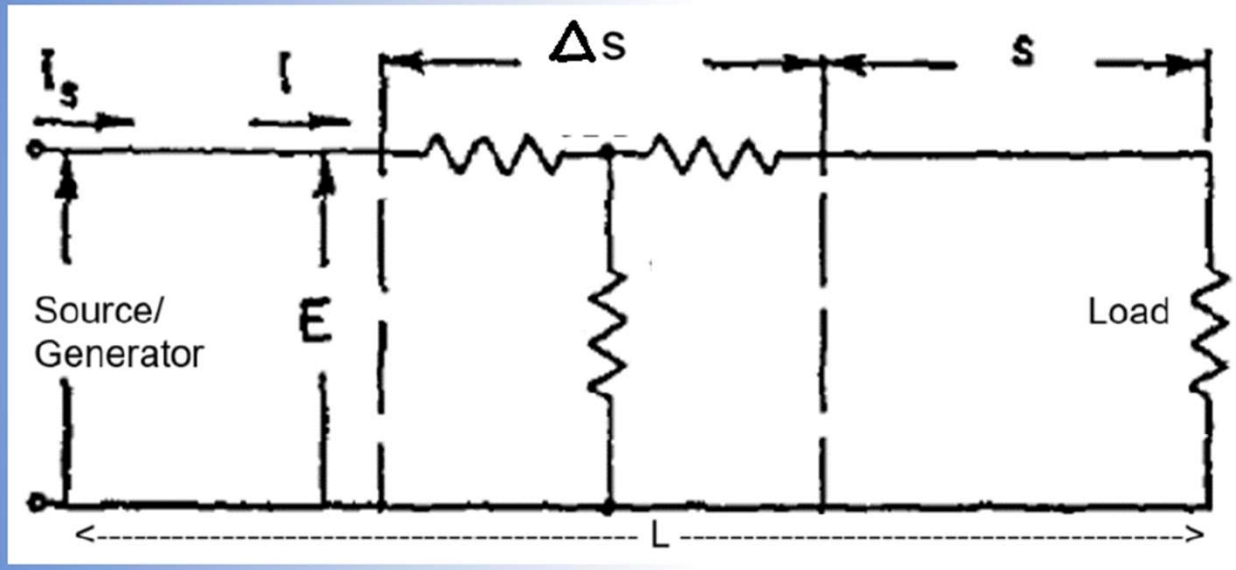
- For f = 10GHz

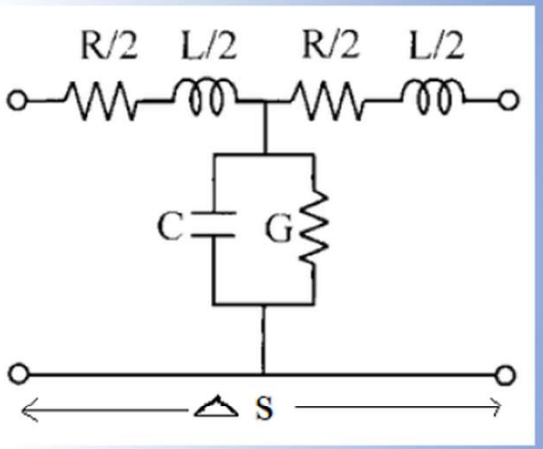
$\lambda = 3cm$



- Note: For a TL, if the parameters are distributed along the length of the conductor – distributed parameter

Lumped parameters:



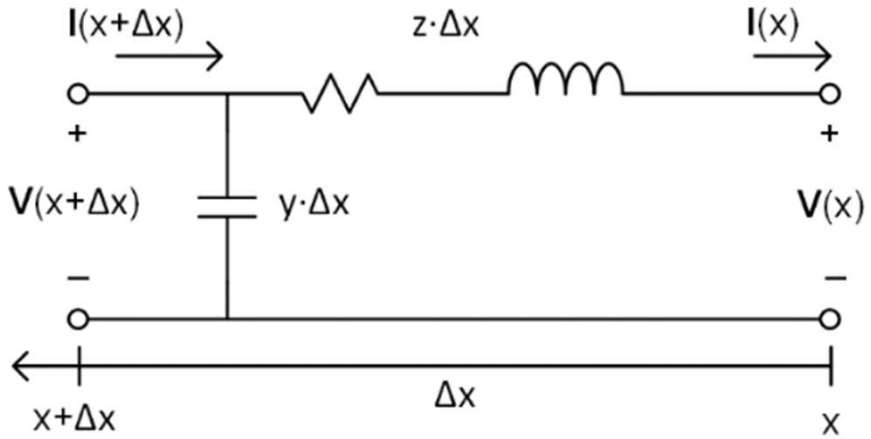


R → Ω/km
L→ mH/km
C→ μF/km
G→ μ℧/km (μS/m)

Electrical Properties of Transmission Lines

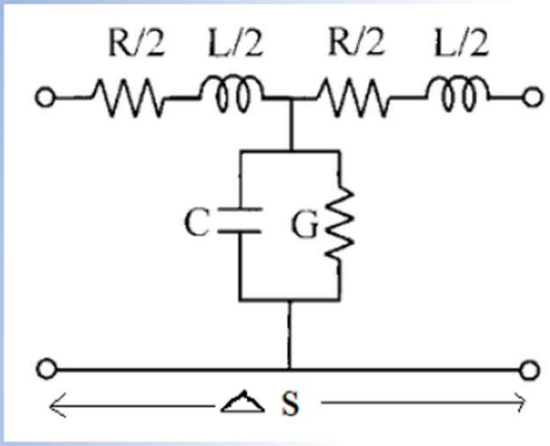
- ***Series resistance***
  - Voltage drop (IR) and real power loss (I^2R) along the line
  - Due to finite conductivity of the line
- ***Series inductance***
  - Series voltage drop, no real power loss
  - Only self inductance (no mutual inductance) in balanced systems
- ***Shunt conductance***
  - Real power loss (V^2G)
  - Leakage current due to corona effects or leakage at insulators
  - Typically neglected for overhead lines
- ***Shunt capacitance***
  - Capacitance to other conductors and to ground
  - Line-charging currents

- The medium- and short-line models are ***lumped*** models
  - All series impedance lumped into one element
  - Shunt admittances lumped into two elements
- Real lines are ***distributed*** networks
  - Lumped models are inaccurate for long lines
- To treat a line as a distributed network, consider the impedance and admittance of a segment of ***differential length***, Δx



T section:

Characteristic impedance (Z_o) and
Propagation constant (γ)



Series arm impedance $Z = R+j\omega L$

Shunt arm impedance $Y = G+j\omega C$

Note: $v = \frac{\omega}{\beta}$

$\beta = \frac{\omega}{v} = \frac{2\pi f}{f\lambda} = \frac{2\pi}{\lambda}$

Characteristic impedance (Z_o) = $\sqrt{\frac{Z}{Y}}$

Propagation constant (γ) = $\sqrt{ZY}$ = $\alpha+j\beta$

α = attenuation constant (dB/km or Np/km); 1Neper = 8.686 dB

β = Phase constant (rad/km)

Problem Statement

Given:

$$R = 2\ \Omega/\text{m}, \quad G = 0.5\ \text{mS/m}, \quad L = 8\ \text{nH/m}, \quad C = 0.23\ \text{pF/m}, \quad f = 1\ \text{GHz}$$

Find:

1. The **Characteristic Impedance** Z_0
2. The **Propagation Constant** γ

All parameters are per unit length (m).

Solution: Characteristic Impedance

Formula:

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Substitute:

$$Z_0 = \sqrt{\frac{2 + j2\pi(10^9)(8 \times 10^{-9})}{0.5 \times 10^{-3} + j2\pi(10^9)(0.23 \times 10^{-12})}}$$

Simplify:

$$Z_0 = 181.39\angle 8.4^\circ = 179.44 + j26.5\ \Omega$$

Hence, the characteristic impedance has a small reactive component.

Solution: Propagation Constant

Formula:

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

Substitute:

$$\gamma = \sqrt{(50.31\angle 87.72^\circ)(15.29 \times 10^{-14}\angle 70.91^\circ)}$$

Simplify:

$$\gamma = 0.2774\angle 79.31^\circ$$

$$\Rightarrow \gamma = 0.051 + j0.273 \text{ per meter}$$

Where: $\alpha = 0.051$ Np/m, $\beta = 0.273$ rad/m

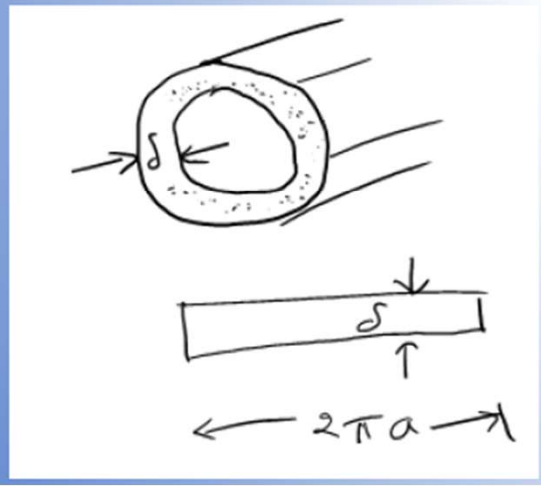
Assumptions made for simplified analysis of TL parameters

1. Skin effect:

   $\omega L >> R$ while computing Z

   <u>Reason:</u> R increases because of skin effect with $\sqrt{f}$ $\quad R_{ac} \propto \sqrt{f}$

   where as line reactance increases directly with f $\quad X_L \propto f$



$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

$$R = \frac{\rho L}{2\pi a \delta}$$

$$R_{ac} = \frac{\rho L \sqrt{\pi f \mu \sigma}}{2\pi a}$$

$$\Rightarrow R_{ac} \propto \sqrt{f}$$

2. G is approximately zero – well constructed
3. Zero dissipation line → R is approximately zero
   (Transmission of power at high efficiency)
4. R < 0 (But not equal to zero)
   (When lines are employed as circuit elements or where resonance properties are involved)

Assumption for Simplified Analysis

The Low-Loss Condition

For a simplified analysis of Transmission Line (TL) parameters, especially the characteristic impedance (Z_0) and phase velocity (V_{ph}), we assume:

$$\omega L \gg R$$

- At sufficiently high operating frequencies, the series impedance Z is primarily determined by the **reactive** component.

$$Z = R + j\omega L \approx j\omega L$$

- This validates the use of $Z_0 \approx \sqrt{L/C}$ and $V_{ph} \approx 1/\sqrt{LC}$.

Reason: The Skin Effect

The validity of $\omega L \gg R$ is maintained at high frequencies because of the Skin Effect.

- As frequency (f) increases, current crowds to the conductor surface.
- The effective cross-sectional area decreases.
- The AC Resistance (R_{ac}) increases with the square root of frequency:

$$R_{ac} \propto \sqrt{f}$$

- Line Reactance ($X_L = \omega L$) increases directly and linearly with frequency:

$$X_L \propto f$$

Conclusion

Since $\boldsymbol{X_L}$ increases faster than $\boldsymbol{R_{ac}}$, at high frequencies ωL quickly dominates R, confirming the initial assumption.

Skin Depth (δ) and AC Resistance (R_{ac})

1. Skin Depth (δ)

The depth from the surface where the current density falls to $1/e$ ($\approx 37\%$) of its surface value:

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

2. AC Resistance (R_{ac})

For a cylindrical conductor, the AC current flows in the conductive shell of thickness δ (Area $\approx 2\pi a\delta$).

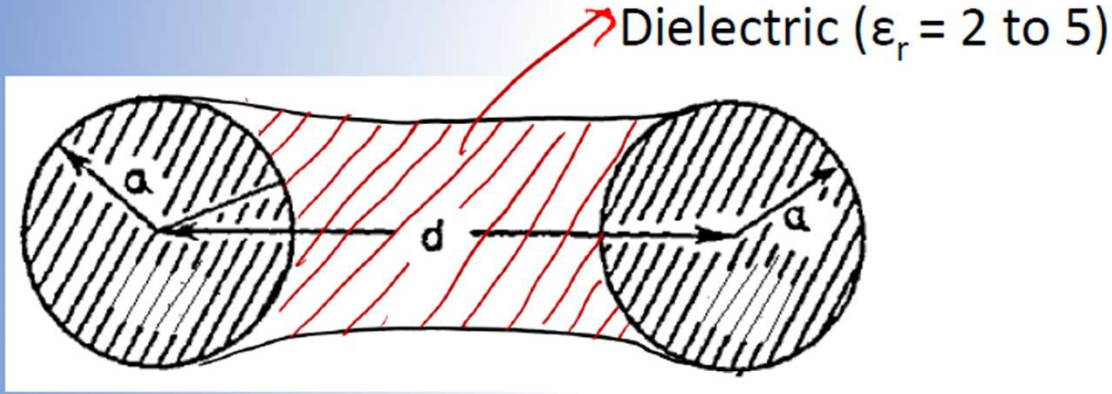
$$R_{ac} = \frac{\rho \times (\text{Length})}{\text{Area}} = \frac{1/\sigma \times \ell}{2\pi a \delta}$$

Substituting the formula for δ:

$$R_{ac} = \frac{\ell}{2\pi a \sigma} \cdot (\sqrt{\pi f \mu \sigma}) \quad \Longrightarrow \quad R_{ac} \propto \sqrt{f}$$

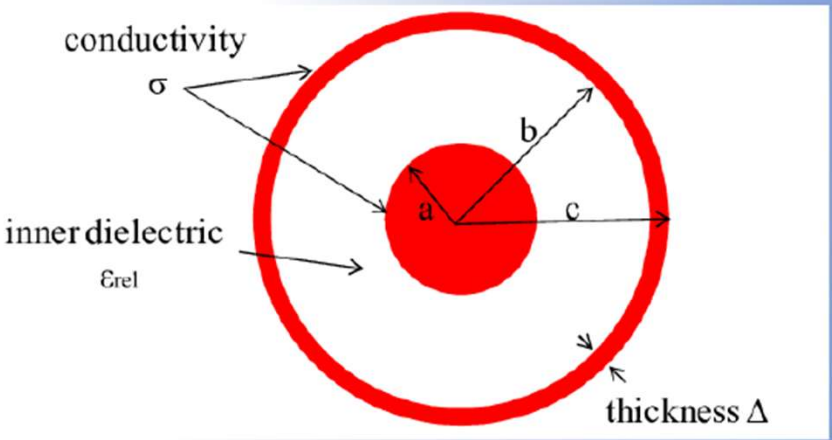
Parameters:

- Flat cable



Ro = 200 – 700 Ω

Coaxial cable



Ro = 40 – 90 Ω

Inductance (L) per unit meter

General Formula:

$$L = \frac{\mu}{\pi} \ln\left(\frac{d}{a}\right) \quad \text{(H/m)}$$

For Non-magnetic Materials ($\mu_r = 1$):

- $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m
- $L = \frac{4\pi \times 10^{-7}}{\pi} \ln\left(\frac{d}{a}\right)$
- $L = 4 \times 10^{-7} \ln\left(\frac{d}{a}\right)$ H/m

Capacitance (C) per unit meter

General Formula:

$$C = \frac{\pi\epsilon}{\ln\left(\frac{d}{a}\right)} \quad \text{(F/m)}$$

With Dielectric ($\epsilon = \epsilon_0\epsilon_r$):

- $\epsilon_0 \approx 8.854 \times 10^{-12}$ F/m
- $C = \frac{\pi(8.854 \times 10^{-12})\epsilon_r}{\ln\left(\frac{d}{a}\right)}$
- $C \approx 27.808 \times 10^{-12} \frac{\epsilon_r}{\ln\left(\frac{d}{a}\right)}$ F/m

Inductance (L) per unit meter

General Formula:

$$L = \frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right) \quad \text{(H/m)}$$

For Non-magnetic Dielectric ($\mu_r = 1$):

- $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m
- $L = \frac{4\pi \times 10^{-7}}{2\pi} \ln\left(\frac{b}{a}\right)$
- $L = 2 \times 10^{-7} \ln\left(\frac{b}{a}\right)$ H/m

Capacitance (C) per unit meter

General Formula:

$$C = \frac{2\pi\epsilon}{\ln\left(\frac{b}{a}\right)} \quad \text{(F/m)}$$

With Dielectric ($\epsilon = \epsilon_0\epsilon_r$):

- $\epsilon_0 \approx 8.854 \times 10^{-12}$ F/m
- $C = \frac{2\pi(8.854\times10^{-12})\epsilon_r}{\ln\left(\frac{b}{a}\right)}$
- $C \approx 55.63 \times 10^{-12} \frac{\epsilon_r}{\ln\left(\frac{b}{a}\right)}$ F/m

General Characteristic Impedance Formula

The characteristic impedance (Z_0) is derived from the primary parameters (R, L, G, C) and is crucial for wave analysis.

$$Z_0 = \sqrt{\frac{Z}{Y}}$$

Series Impedance (Z)

- Resistance and Inductance per unit length.
- $Z = R + j\omega L$

Shunt Admittance (Y)

- Conductance and Capacitance per unit length.
- $Y = G + j\omega C$

The Low-Loss Approximation

High-Frequency / Low-Loss Assumption

At high frequencies, or for lines with very small loss components, the reactive terms dominate the primary parameters:

- $R \ll \omega L \implies Z \approx j\omega L$
- $G \ll \omega C \implies Y \approx j\omega C$

Simplified Characteristic Impedance (R_0)

Substituting the approximations back into Z_0:

$$Z_0 \approx \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}} \quad [\Omega]$$

- This resulting impedance (R_0) is purely **resistive** and **frequency-independent**.

Finalized Formula: Flat/Parallel-Wire Cable

- The low-loss impedance $R_0 = \sqrt{L/C}$ is determined solely by the line's geometry and the dielectric constant (ϵ_r).
- The general formula is then resolved into a function of these physical dimensions.

The Finalized Formula

$$R_0 = \frac{120}{\sqrt{\epsilon_r}} \ln\left(\frac{d}{a}\right) \quad [\Omega]$$

Parameter Definitions

- ϵ_r: Relative permittivity (dielectric constant)
- d: Center-to-center spacing of conductors
- a: Radius or equivalent size of conductors

- This formula is the final, practical result for calculating the characteristic impedance of a two-wire parallel line.

Finalized Formula for Coaxial Cable

Final Simplification

Substituting L and C into $R_0 = \sqrt{L/C}$ and performing algebraic cancellation yields the simplified, practical formula:

$$R_0 = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right) \qquad [\Omega]$$

The Resulting Constants:

- **Attenuation Constant (α):** $\alpha = 0$
- **Phase Constant (β):** $\beta = \omega\sqrt{LC}$

The Propagation Constant (γ)

General Formula

The propagation constant governs how a wave's amplitude and phase change along the line:

$$\gamma = \sqrt{ZY} = \alpha + j\beta$$

Lossless Line Approximation ($R = 0, G = 0$)

Under the lossless condition, $Z = j\omega L$ and $Y = j\omega C$:

$$\gamma = \sqrt{(j\omega L)(j\omega C)}$$

$$\gamma = \sqrt{-\omega^2 LC} = j\omega\sqrt{LC}$$

Derivation of V_{ph} for Flat Cable (Parallel Wires)

Starting Formula

$$V_{ph} = \frac{1}{\sqrt{LC}}$$

Substituting L and C for Flat Cable (d: separation, a: conductor radius):

- $L_{\text{flat}} = \frac{\mu_0 \mu_r}{\pi} \ln\left(\frac{d}{a}\right)$
- $C_{\text{flat}} = \frac{\pi \epsilon_0 \epsilon_r}{\ln\left(\frac{d}{a}\right)}$

The Geometric Cancellation Step

$$V_{ph} = \frac{1}{\sqrt{\left(\frac{\mu_0 \mu_r}{\pi} \ln\left(\frac{d}{a}\right)\right)\left(\frac{\pi \epsilon_0 \epsilon_r}{\ln\left(\frac{d}{a}\right)}\right)}}$$

$$V_{ph} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} \quad \text{(Geometric terms cancel)}$$

Phase Velocity (V_{ph}) Final Universal Result

Relating to the Speed of Light (c)

The speed of light in vacuum is: $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

The Final Formula:

- Assuming $\mu_r \approx 1$, the formula simplifies to:

$$V_{ph} = \frac{c}{\sqrt{\epsilon_r}} \quad [\mathrm{m\,s^{-1}}]$$

- This is a universal result for **any** TEM transmission line (coaxial, flat, etc.) under the lossless assumption.
- The velocity is only reduced by the dielectric constant (ϵ_r).

TEM stands for **Transverse Electromagnetic.**

TEM	$E_z = 0, H_z = 0$	Coaxial, Flat Cable

Problem Statement

An air-filled coaxial line of copper is to have a capacitance of $C = 22\,\text{pF m}^{-1}$. The inner conductor has a diameter of $0.1\,\text{cm}$. Calculate:

1. The inductance (L) of the line.
2. The inner radius of the outer conductor (b).
3. Characteristic Impedance (R_0).
4. The phase constant (β) and wavelength (λ) at a frequency of $f = 25\,\text{MHz}$, neglecting dissipation.

Given & Derived Parameters

- **Capacitance (C):** $22 \times 10^{-12}\,\text{F m}^{-1}$
- **Frequency (f):** $25 \times 10^{6}\,\text{Hz}$
- **Inner Radius (a):** $\frac{0.1\,\text{cm}}{2} = 0.05 \times 10^{-2}\,\text{m}$
- **Dielectric Constant (ϵ_r):** 1 (Air-filled)

Solution: Geometry and Inductance (b and L)

We use the numerical formula for capacitance, $C = \frac{55.63\epsilon_r}{\ln(b/a)}$ [pF/m]

$$\ln\left(\frac{b}{a}\right) = \frac{55.63 \times 1}{C\ [\text{pF/m}]}$$

$$\ln\left(\frac{b}{a}\right) = \frac{55.63}{22} \approx \mathbf{2.5286}$$

The geometric ratio is $b/a = e^{2.5286} \approx \mathbf{12.535}$.

$$b = 12.535 \times a = 12.535 \times 0.05 \times 10^{-2}\,\text{m} = 6.268 \times 10^{-3}\,\text{m}$$

Using the ratio $\ln(b/a)$ found above in the numerical formula:

$$L = 0.2 \ln\left(\frac{b}{a}\right) \quad [\mu\text{H/m}]$$

$$L = 0.2 \times 2.5286 = 0.5057\,\mu\text{H}\,\text{m}^{-1}$$

Solution: Impedance and Propagation (R_0 and β)

We use the simplified formula for R_0 for a lossless line:

$$R_0 = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right) \quad [\Omega]$$

$$R_0 = \frac{60}{\sqrt{1}} \times 2.5286 = 151.72$$

For a lossless line, the phase constant is $\beta = \omega\sqrt{LC}$:

$$\beta = 2\pi f\sqrt{LC}$$
$$\beta = 2\pi(25 \times 10^6\,\text{Hz})\sqrt{(0.5057 \times 10^{-6}\,\text{H m}^{-1})(22 \times 10^{-12}\,\text{F m}^{-1})}$$
$$\beta = 2\pi(25 \times 10^6\,\text{Hz})(3.335 \times 10^{-9})$$
$$\beta \approx 0.5236\,\text{rad m}^{-1}$$

Solution: Wavelength (λ)

We can calculate the wavelength using two methods:

Method 1: Using β

The wavelength is $\lambda = 2\pi/\beta$.

$$\lambda = \frac{2\pi}{0.5236\,\text{rad m}^{-1}}$$

$$\lambda = 12.0\,\text{m}$$

Method 2: Using Phase Velocity (V_{ph})

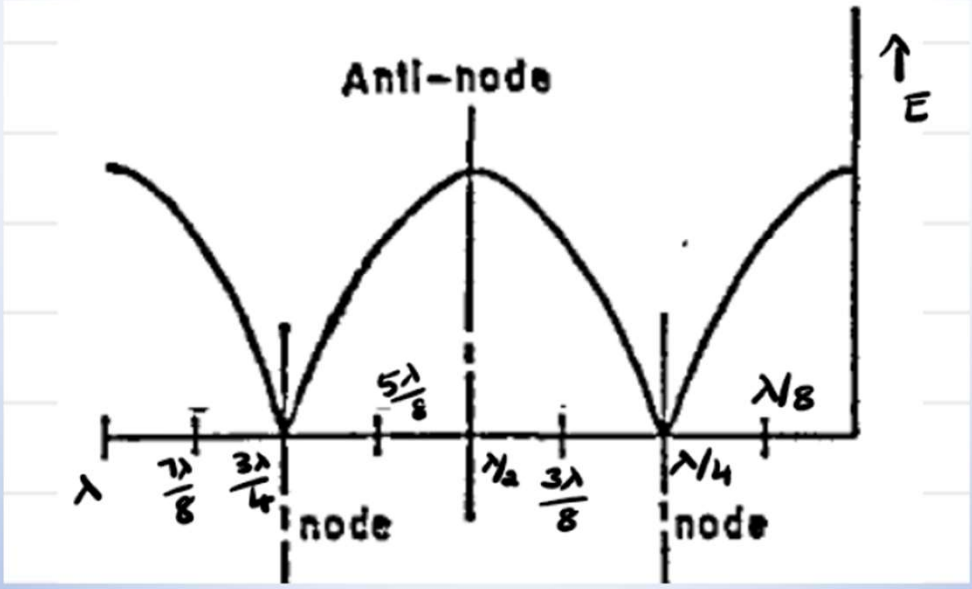
For air-filled lines, $V_{ph} = c$.

$$\lambda = \frac{V_{ph}}{f} = \frac{c}{f}$$

$$\lambda = \frac{3 \times 10^8\,\text{m s}^{-1}}{25 \times 10^6\,\text{Hz}}$$

$$\lambda = 12.0\,\text{m}$$

- If the resultant total voltage appears to be standstill, oscillating with in magnitude with time but having fixed positions of maxima and minima
- Such a wave is known as a standing wave
- If the line voltages are measured with an RMS voltmeter, the magnitude will appear as shown below.
- Note: Voltmeter doesn't distinguish between +ve and –ve values



- $S = \frac{|E_{max}|}{|E_{min}|} = \frac{1 + |K|}{1 - |K|}$ OR $|K| = \frac{s - 1}{s + 1}$
- This relation may be rearranged as

$$|K| = \frac{|E_{max}| - |E_{min}|}{|E_{max}| + |E_{min}|}$$

$|K| = |Er|/|Ei| = 1$ if $|Er| = |Ei|$

For $|K| = 0$, $S = 1$

For $|K| = 1$, $S = \infty$

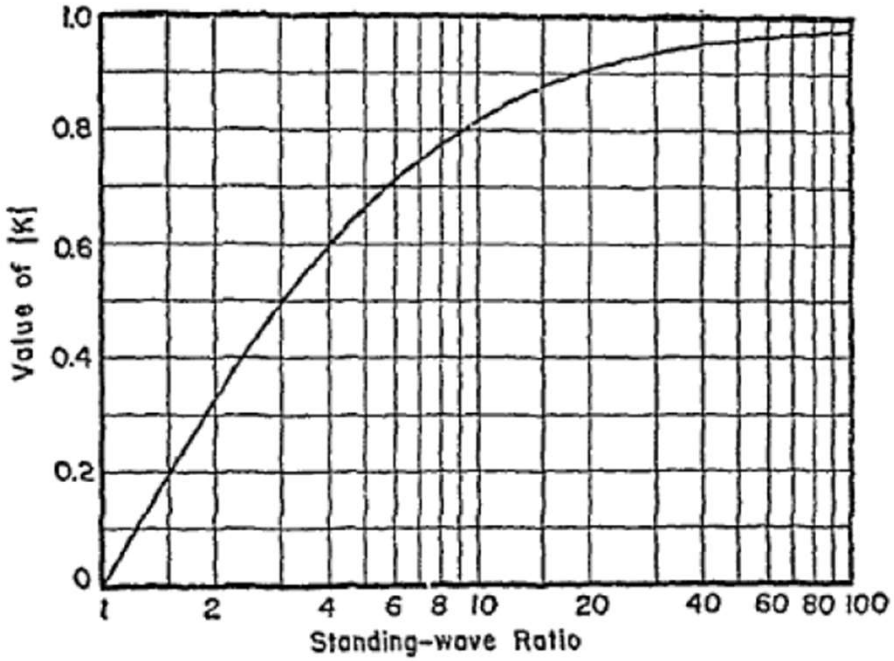
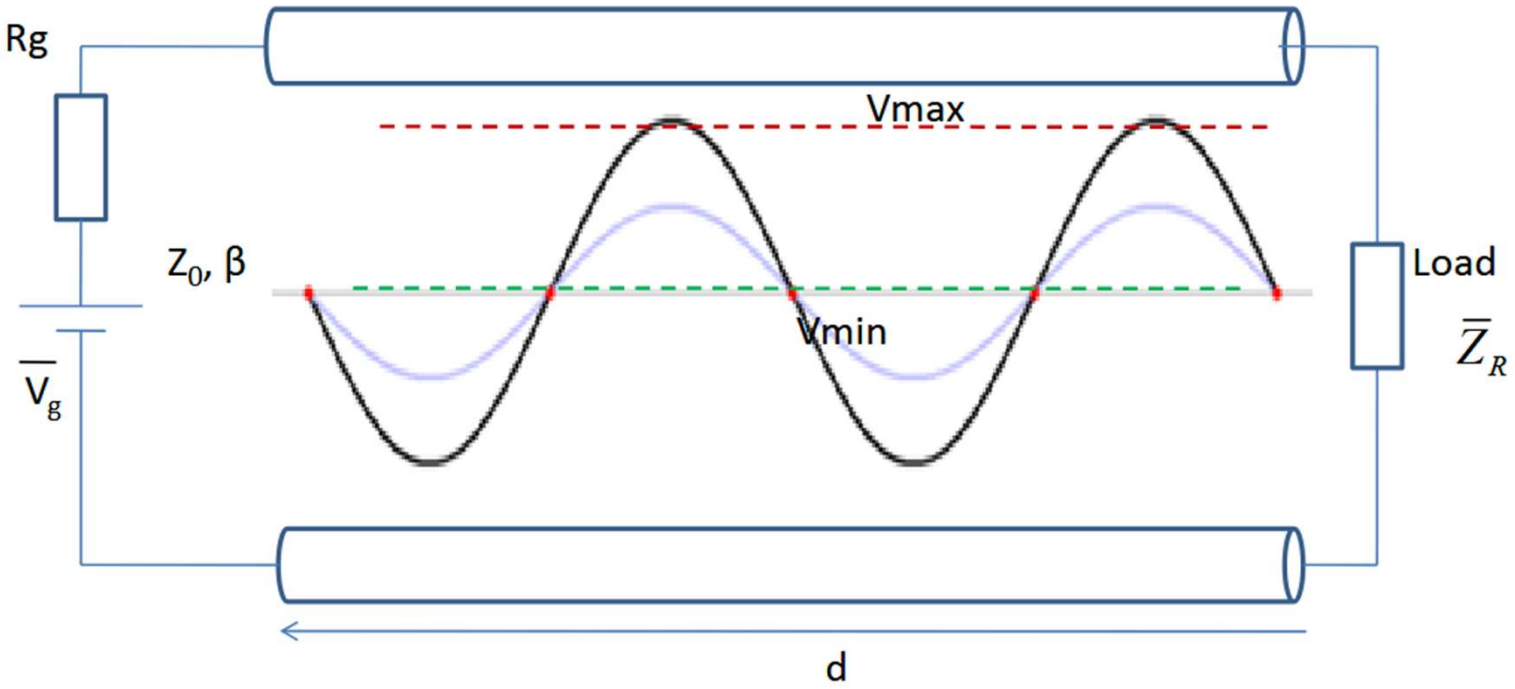


Fig. 7-8. Relation between the standing-wave ratio S and the magnitude of the reflection coefficient.

Standing Wave Ratio



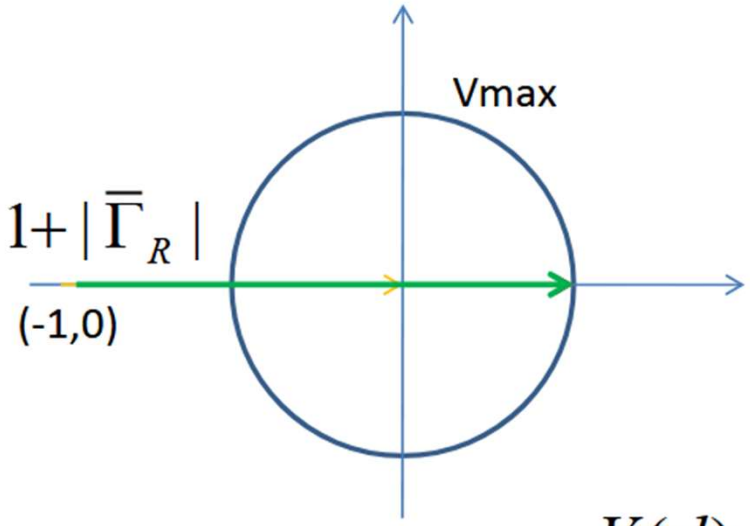
[Pg.452, Eqn. 7.25a]

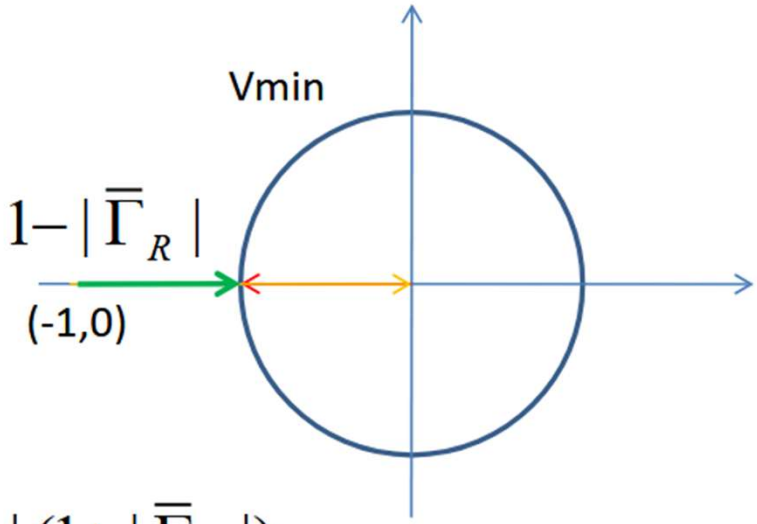
$$|V(d)| = |V^+||1+\Gamma_R e^{-j2\beta d}|$$

[Pg.454, Eqn. 7.26]

$$SWR = \frac{V(d)_{\max}}{V(d)_{\min}} = \frac{|V^+|(1+|\overline{\Gamma}_R|)}{|V^+|(1-|\overline{\Gamma}_R|)}$$

Meaning of SWR





$$SWR = \frac{V(d)_{\max}}{V(d)_{\min}} = \frac{|V^+|(1+|\overline{\Gamma}_R|)}{|V^+|(1-|\overline{\Gamma}_R|)}$$

1. Short circuit line $\overline{\Gamma}_R = -1 \Rightarrow SWR \to \infty$
2. open circuit line $\overline{\Gamma}_R = 1 \Rightarrow SWR \to \infty$

Complete standing wave

3. Terminated with Z_0 $\overline{\Gamma}_R = 0 \Rightarrow SWR = 1$

NO standing wave